

STATE-DIVERGENCE BUGS ARE FAILED DISTRIBUTIVE LAWS RE-ENTRANCY, STATE DESYNCHRONISATION, AND THE DEGREE-TWO OBSTRUCTION TO COMPOSING CONTRACTS

MACBETH

ABSTRACT. A recurring and expensive class of smart-contract exploit arises not from a faulty component but from the *unsound composition* of two individually-correct ones. Security auditors catalogue these under loose, overlapping headings — “re-entrancy”, “state desynchronisation”, “callback-ordering” — with no common invariant. We argue that they share a precise mathematical shape. An on-chain effect that issues calls and receives replies is a container (polynomial functor); a handoff discipline on such effects is a directed container, equivalently a small category; and interleaving two shared-state effects is their Zappa–Szép product $\mathcal{C} \bowtie \mathcal{D}$, which exists exactly when a distributive law of directed containers does. Its non-existence is a degree-two cohomology class $[\omega] \in H^2$. Our anchor is a theorem, with a machine-checked core: on the minimal faithful supervisor–worker model, unprotected re-entrancy is the nonzero class $[\omega] = \varepsilon$ in $H^2 \cong \mathbb{F}_2$, and the joint contract exists if and only if $\varepsilon = 0$. We then advance, explicitly as a *modelling claim rather than a theorem*, that the state-desynchronisation family — illustrated by the 2026 Provenance Blockchain marker/bank bug and three of Trail of Bits’ Uniswap-v4 hook failure patterns — is the same obstruction for different effects, and we set out the programme required to turn that claim into a theorem.

1. INTRODUCTION

A bug that belongs to no single component. The most costly smart-contract failures are often not bugs *in* a component but bugs *between* components. The 2016 DAO exploit drained roughly a third of the Ethereum DAO by re-entrancy: a contract called an external party before updating its own ledger, and the callee re-entered the open call and spent against stale state. Each line of the contract was locally defensible; the failure lived in the interaction. The same signature recurs across unrelated systems. In August 2026 Trail of Bits disclosed a Provenance Blockchain vulnerability in which a token’s *marker* module and *bank* module each maintained a representation of circulating supply; the two were individually sound, but their divergence let an authorization check degenerate to $0 = 0$ and granted an attacker free mint and withdraw rights [11]. In July 2026 the same auditors catalogued seven recurring failure modes for Uniswap-v4 hooks — user callbacks spliced into the core swap lifecycle — several of which are, in our reading, two locally-correct pieces of accounting that disagree once composed, including the \$8.4M Bunni exploit [12].

Auditors group these under headings that overlap and shade into one another: re-entrancy, state desynchronisation, check-then-act races, callback-ordering. The practical question this note takes up is whether the grouping reflects a *mathematical* commonality — an invariant that is zero exactly when the composition is safe and nonzero exactly when it is not, and which can be *computed* rather than argued by analogy.

The proved anchor. For one member of the family the answer is yes, and it is a theorem. We recall it in Section 3; here is the shape. An on-chain effect that may issue queries from a set S and, for each query s , receive a reply from $P(s)$, is the container $p = (S, P)$, i.e. the polynomial functor $\sum_{s \in S} y^{P(s)}$ [1]. A discipline governing who may hand off to whom is a *directed container*, which by Ahman–Uustalu is exactly a small category [3]. Running two effects that share a resource is not

composition along one fixed discipline — a functor already models that, and a functor cannot fail — but the *interleaving* of two, and the interleaving of two directed containers is their *Zappa–Szép product* $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$, assembled from a distributive law of directed containers [4]. Such a product can fail to exist, and (under mild hypotheses) its existence is governed by a single degree-two class,

$$\mathcal{C} \bowtie \mathcal{D} \text{ exists} \iff [\omega] = 0 \in H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D}),$$

the Baues–Wirsching cohomology of the handoff category’s skeleton with coefficients in the reply functor [8, 7, 6]. On the minimal faithful supervisor–worker model $\mathcal{K}_{\varepsilon}$, parametrised by a one-bit turn token whose mutation bit ε records whether a worker writes to the supervisor’s shared state during a pending call, the class is the bit on the nose: $[\omega] = \varepsilon$ in $H^2 \cong \mathbb{F}_2$, so the joint contract exists if and only if $\varepsilon = 0$ [9]. The finite-field core of this identity is machine-checked in Lean 4, sorry-free and with no external library [10].

The thesis of this note. Our contribution here is not a new theorem but a *frame*, advanced honestly as such. We propose that the state-desynchronisation family is the *same obstruction for different effects*: two on-chain effects that each maintain a representation of a shared logical value compose soundly iff a distributive law forces their representations to agree, and the catalogued bugs are exactly the cases where no such law exists — where the defect cocycle of every transversal is nonzero. We make this precise enough to be falsifiable, we work it through on the Provenance and Uniswap-v4 examples, and — crucially — we mark the boundary between what is proved and what is modelled. *The re-entrancy anchor is a theorem with a machine-checked core. The extension to state desynchronisation is a modelling claim: a dictionary and a worked mapping, graded as such, together with the explicit list of what one would have to prove to upgrade it.*

Why this matters for impact. Compositional correctness here has a direct economic denominator: the failures above cost real funds, and they are failures of composition, not of components. A framework that *predicts* the bug class from the absence of a distributive law — that says “these two effects cannot be safely interleaved, and here is the class that certifies it” — turns an open-ended audit for “smells” into the computation of a specific invariant. That is the promise this note stakes out; Section 6 is candid about how much of it is currently cashed.

Outline. Section 2 fixes the container / directed-container / Zappa–Szép vocabulary and the two-level existence criterion. Section 3 states the re-entrancy theorem and draws its verification boundary honestly. Section 4 isolates the structural signature — two lawful factors, unsound composite — as the failure of condition (G). Section 5 presents the state-desynchronisation family as a second instance, explicitly as a modelling claim, and enumerates the obligations a theorem would discharge. Section 6 states the grant impact and the research programme. Section 7 closes.

2. PRELIMINARIES

We include only what the statement and the modelling dictionary use. Containers, the directed-container / small-category identification, and the Zappa–Szép construction are prior art and are cited as such.

Containers and directed containers. A *container* $p = (S, P)$ is a set S of *shapes* together with, for each $s \in S$, a set $P(s)$ of *positions*; it presents the polynomial functor $\sum_{s \in S} y^{P(s)}$ [1, 2]. Read as an interactive effect, a shape is a query the effect may issue and a position is a reply it may receive. A *directed container* equips (S, P) with a root, a sub-position operation, and a position concatenation; Ahman–Uustalu’s theorem is that directed containers are *exactly* small categories — objects are shapes, morphisms out of s are the positions $P(s)$, identities are roots, composition is concatenation [3]. We use the category picture throughout and call it the *handoff category*: its morphisms are the legal ways control and state pass between effects.

Zappa–Szép products and distributive laws. Let $\mathcal{K}$ be a small category and $\mathcal{D} \subseteq \mathcal{K}$ a *wide* subcategory (all objects, some morphisms). A *Zappa–Szép product* $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ is a second wide subcategory $\mathcal{C}$ such that every morphism of $\mathcal{K}$ factors *uniquely* as a $\mathcal{C}$ -morphism followed by a $\mathcal{D}$ -morphism: a strict factorization system, equivalently a distributive law $\lambda : \mathcal{D}\mathcal{C} \Rightarrow \mathcal{C}\mathcal{D}$ satisfying the matched-pair axioms [5]. For directed containers this is the composition construction of Ahman–Uustalu [4]: it generalises the Zappa–Szép product of two monoids, and its data is a matched pair of mutual actions — the two factors acting on one another. The point for us is that, given $\mathcal{D}$, the complementary factor $\mathcal{C}$ *may or may not exist*; that existence is what an obstruction measures.

Two effects that each operate correctly in isolation are the two *factors*. Interleaving them on shared state is asking for the product $\mathcal{C} \bowtie \mathcal{D}$. A distributive law is exactly a coherent rule for commuting one factor past the other — for saying how the state left by one is seen by the other. Its absence is not a malfunction of either factor; it is the absence of a law relating them.

The two-level existence criterion. The product’s existence splits into two conditions [7]. Write $\text{Hom}_{\mathcal{K}}(-, b)$ for the contravariant hom-presheaf of $\mathcal{K}$.

- (L) (*freeness*) each $\text{Hom}_{\mathcal{K}}(-, b)$ is a free right $\mathcal{D}$ -set, i.e. a coproduct of representables;
- (G) (*global closure*) the free bases close into a wide subcategory $\mathcal{C}$.

Theorem 2.1 (pairwise Zappa–Szép criterion [7, 5]). *For a small category $\mathcal{K}$ and a wide subcategory $\mathcal{D} \subseteq \mathcal{K}$, a strict factorization $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ exists if and only if (L) and (G) both hold.*

Condition (L) is local: each hom-presheaf, considered alone, is free. Condition (G) is global: the chosen bases must assemble consistently. The split is exactly the signature we will exploit — (L) is the two factors being individually well-behaved; (G) is their interleaving being coherent.

Theorem 2.2 (the global obstruction is cohomological [8, 6]). *Assume (L), and the hypothesis (H) that the reply (direction) functor $\mathcal{D}$ is a disjoint union of abelian vertex groups with trivial left action. Then the free $\mathcal{D}$ -orbits form the skeleton category $\text{Sk}_{\mathcal{C}}$, the vertex groups form a presheaf $\mathcal{D} : \text{Sk}_{\mathcal{C}}^{\text{op}} \rightarrow \mathbf{Ab}$, and for any transversal T the defect 2-cochain ω_T , defined by $c_2 \circ c_1 = [c_2 c_1]_T \circ \omega_T(c_2, c_1)$, is a normalized 2-cocycle whose class $[\omega] \in H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D})$ is transversal-independent, with*

$$(G) \iff [\omega] = 0.$$

The obstruction is the *class* $[\omega]$, not the cochain ω : a transversal may have $\omega_T \neq 0$ while $[\omega] = 0$, in which case closure still exists. This is the categorical form of “a flat bundle is trivial iff its holonomy vanishes”: each partial handoff completes, but the completions may fail to glue into a single global system by a monodromy around a loop.

3. THE PROVED ANCHOR: RE-ENTRANCY IS A FAILED DISTRIBUTIVE LAW

This section restates a shipped result [9, 10]; we do not re-prove it. We state it precisely because it is the one rung of the ladder that is a theorem, and because its verification boundary is the template for grading everything that follows.

The minimal faithful model. Fix three roles: S (supervisor), W (a worker running), R (a return point). The supervisor carries a one-bit *turn token*; toggling it is an involution τ , so $\text{End}(S) = \{\text{id}_S, \tau\} \cong \mathbb{Z}/2$. The generating handoffs are dispatches $p, p\tau : S \rightarrow W$ and two worker outcomes $s, s_2 : W \rightarrow R$, with composites $q, q\tau : S \rightarrow R$. The prescribed right factor is the supervisor-internal (token) part $\mathcal{D} = \{\text{id}_S, \tau, \text{id}_W, \text{id}_R\}$; the left factor $\mathcal{C}$ — the genuine dispatch/return part — is what a Zappa–Szép product must supply.

Definition 3.1 (the family $\mathcal{K}_{\varepsilon}$ [9]). For $\varepsilon \in \mathbb{Z}/2$ let $\mathcal{K}_{\varepsilon}$ be the above category with the single length-two composite fixed by

$$s \circ p = q, \quad s_2 \circ p = q \cdot \tau^{\varepsilon},$$

where s is “the worker returns normally” and s_2 is “the worker re-enters” — calls back into the supervisor before returning. Thus $\varepsilon = [s_2 \circ p = q\tau]$ records whether the re-entrant outcome *mutates* the shared token ($\varepsilon = 1$, unprotected re-entry) or *fixes* it ($\varepsilon = 0$, a lock).

The theorem. The reduction of [9] verifies (L) and (H) for $\mathcal{K}_\varepsilon$, identifies the skeleton $\text{Sk}_\mathcal{C}$ as the branch $S \rightarrow W \rightrightarrows R$ with $[s][p] = [s_2][p] = [q]$, and computes the Baues–Wirsching complex to be $C^1 \cong \mathbb{F}_2^2$, $C^2 \cong \mathbb{F}_2^2$, $C^3 = 0$, with coboundary $\delta^1 h = (h[p] - h[q], h[p] - h[q])$ — so $B^2 = \langle (1, 1) \rangle$ is the diagonal and $H^2 = \mathbb{F}_2^2 / B^2 \cong \mathbb{F}_2$ — and defect cocycle $\omega_T(\varepsilon) = (0, \varepsilon)$.

Theorem 3.2 (re-entrancy obstruction [9]). *On the minimal faithful model, the obstruction class equals the token-mutation bit,*

$$[\omega(\mathcal{K}_\varepsilon)] = \varepsilon \in H^2(\text{Sk}_\mathcal{C}; \mathcal{D}) \cong \mathbb{F}_2,$$

and hence the joint contract $\mathcal{C} \bowtie \mathcal{D}$ exists if and only if $\varepsilon = 0$. Unprotected re-entrancy ($\varepsilon = 1$) is the nonzero generator of $H^2 \cong \mathbb{Z}/2$.

The verification boundary, drawn honestly. The finite-field core of Theorem 3.2 is machine-checked in Lean 4 (`Containers/Reentrancy.lean`, sorry-free, no external library [10]). Modelling $\mathbb{F}_2$ as `Bool` with `xor` as addition, the class map $\varphi(a, b) = a \oplus b$, the coboundary `d1 t = (t, t)`, and the defect `omega ε = (0, ε)`, the file certifies $\ker \varphi = B^2$ and φ surjective (so $H^2 \cong \mathbb{F}_2$ via φ), $\varphi(\omega_\varepsilon) = \varepsilon$ (the class form of $[\omega] = \varepsilon$), and $\omega_\varepsilon \in B^2 \iff \varepsilon = 0$ (the membership form of the dichotomy). What Lean checks is the *last mile*: from the cochain complex to $[\omega] = \varepsilon$. What it does *not* check is the *first mile*: that $\mathcal{K}_\varepsilon$ satisfies (L) and (H), that $\text{Sk}_\mathcal{C}$ is the stated branch, and that the transversal defect is $(0, \varepsilon)$. Those are established by hand in [9] and imported here as a cited, pen-and-paper input. We state this boundary now because it is exactly the kind of boundary the next two sections must respect: the honest unit of currency is “theorem with machine-checked core”, and anything short of it is graded down.

4. THE STRUCTURAL SIGNATURE: TWO LAWFUL FACTORS, AN UNSOUND COMPOSITE

Strip Theorem 3.2 to its shape. Re-entrancy is dangerous not because dispatch is wrong or because the token logic is wrong — each factor is a perfectly good wide subcategory — but because, when $\varepsilon = 1$, the two *cannot be interleaved into one consistent contract*. In the language of Theorem 2.1: condition (L) holds (the factors are individually free), and condition (G) fails (they do not close). The obstruction $[\omega]$ lives entirely in the gap between (L) and (G).

This gives a precise reading of the situation auditors describe when two components are each correct in isolation yet fail once composed:

(L) *holds* — each component, as an effect on its own state, is a lawful directed container; and (G) *fails* — there is no distributive law interleaving them, so the composite is not a single consistent contract. The exploit is a witness that $[\omega] \neq 0$.

Two points make this more than a slogan. First, it is *falsifiable*: a bug claimed to be of this family must exhibit two factors each satisfying (L), a shared resource over which they are to be interleaved, and a nonzero class obstructing the interleave; if any component is simply wrong in isolation — (L) fails — it is an ordinary bug, not a composition bug, and the framework makes no claim about it. Second, it is *computational*: when the handoff category and reply functor are pinned down and (H) holds, $[\omega]$ is a finite linear-algebra computation of the kind Lean already discharges for the anchor. The content of the sections below is the claim that the state-desynchronisation family fills in this template — and an honest accounting of how much of the template is currently filled.

5. STATE DESYNCHRONISATION AS A SECOND INSTANCE

We now present the state-desynchronisation family as a second instance of the signature of Section 4. We stress at the outset:

Modelling claim 5.1. The following is a *modelling claim, not a theorem*. We assert that the state-desynchronisation bug family admits a faithful directed-container model in which the two divergent representations of a shared value are the two factors, condition (L) holds, and the bug is the non-vanishing of the global-closure class $[\omega]$ of Theorem 2.2. We support it with a dictionary and two worked mappings, and in Section 6 we list the obligations required to make it a theorem. No step below is claimed proved.

The dictionary. The mapping from audit vocabulary to the categorical invariant is as follows.

Audit notion	Container model	Status
effect maintaining a value	directed container (handoff category)	established [3]
two reps. of one logical value	two factors $\mathcal{C}, \mathcal{D}$ over a shared object	modelling
each component correct alone	condition (L)	modelling
“they diverge when composed”	defect cocycle $\omega_T \neq 0$	modelling
no safe interleaving exists	condition (G) fails, $[\omega] \neq 0$	modelling
the exploit transaction	a witness that $[\omega] \neq 0$	modelling
a consistency invariant that forbids it	a distributive law $\lambda : \mathcal{DC} \Rightarrow \mathcal{CD}$	modelling

The conceptual content of the middle rows is a single idea: a distributive law between two representations is exactly a natural, coherent rule forcing them to agree. “State desynchronisation” is the *absence* of such a rule — nothing in the system’s type structure compels the two representations to track one another, so the composite of two agreeing-in-isolation components can disagree, and an authorization predicate can silently factor through the wrong one.

First mapping: the Provenance marker/bank divergence. Trail of Bits’ August 2026 disclosure is the cleanest small example [11]. On the Provenance Blockchain, two modules each represent a token’s circulating supply: the *marker* module’s stored **supply** field, and the *bank* module’s live supply. For a class of markers the marker field is never updated and stays at its default 0, while the bank value is the true supply. The vulnerable check `accountControlsAllSupply` compared an attacker’s balance to the marker field; with both equal to 0, the comparison $0 = 0$ returned true and granted control of the entire supply — free mint and withdraw. Trail of Bits name the root cause precisely: “*The root cause of this issue was state desynchronisation. The marker struct and the bank module both represent the token supply, but only the bank module is kept current for non-fixed markers*” [11].

The modelling reading: the marker-module effect and the bank-module effect are the two factors, each sound on its own state. “Supply” is the shared logical value. A distributive law would be a coherent rule making every marker-side read of supply agree with the bank-side value; no such law is present, so the two representations are free to diverge, and the authorization predicate composes with the stale factor rather than the live one. The exploit transaction is a witness that the two representations cannot be interleaved into one consistent notion of supply — the (G)-failure. Note how the auditors’ own remediation principle — “*an authorization predicate must never be satisfiable from the attacker’s default state*” [11] — is, in this reading, the demand that the predicate factor through a representation on which the distributive law holds.

Remark 5.2 (what is and is not asserted here). We do *not* compute an H^2 class for the Provenance system, and we do not claim one is nonzero; Modelling Claim 5.1 asserts that a faithful model *exists* in which the bug is such a class, and the computation is exactly the obligation listed in Section 6. The value of the example is that the source’s own language — two modules representing one value,

kept in sync by nothing — is the failed-distributive-law picture stated in security-audit prose, with the categorical content supplied entirely by us.

Second mapping: Uniswap-v4 hook patterns. Trail of Bits’ July 2026 catalogue of seven Uniswap-v4 hook failure modes [12] supplies richer, higher-stakes instances. User hooks are callbacks spliced into the pool’s swap lifecycle; the pool and the hook each maintain accounting state, and three of the seven patterns are composition failures of the signature of Section 4.

- **Pattern 3 (custom accounting leaks value).** The \$8.4M Bunni exploit of September 2025 was, in the auditors’ words, “*a rounding bug in BunniHook’s idle-balance accounting*”, in which “*each transaction satisfied the PoolManager’s settlement invariant because the bug was in BunniHook’s internal accounting*” [12]. Two representations of the same balances — the PoolManager’s settled deltas and the hook’s internal ledger — each locally invariant, diverge: a textbook two-factor divergence.
- **Pattern 4 (right logic, wrong hook).** “*The beforeSwap hook executes with pre-swap state, but the afterSwap hook sees post-swap state*” [12]. The two callbacks are two factors reading a shared pool state at different points; code correct in one is unsafe in the other. The temporal misalignment is exactly a failure of the two factors to commute — the absence of a law relating pre- and post-swap representations.
- **Pattern 7 (state changes during a callback sequence).** “*Values cached between [beforeSwap and afterSwap] are not automatically safe ... Nested actions can therefore change shared hook storage, pool state, balances, or oracle data before the outer callback sequence finishes*” [12]. This is re-entrancy on shared hook state — structurally the anchor of Section 3 with the hook storage in the role of the supervisor token.

Pattern 7 is the closest to a genuine instance of the proved anchor; patterns 3 and 4 are the state-desynchronisation reading. In all three the auditors’ diagnosis — individually-correct accounting that disagrees once composed — is the (L)-holds-(G)-fails split, stated without category theory.

Remark 5.3 (honest scope of the two mappings). These are illustrative mappings, not models verified against formal semantics of the EVM or the Cosmos SDK. The directed-container structure of each effect, the verification of (L), and the computation of $[\omega]$ are not carried out here; the claim is that the shape matches and that the computation is well-posed. Treating either mapping as a proved theorem would be an overclaim of exactly the kind Section 3’s boundary is designed to prevent.

6. IMPACT AND THE RESEARCH PROGRAMME

Why the framing has economic value. The failures above — the DAO, Provenance, Bunni — cost real funds, and they are failures of composition rather than of components. Current practice detects them by pattern-matching against a catalogue of smells; the catalogue is opened precisely because the smells have no common definition. The container framing offers a single invariant in their place: two effects over a shared resource compose soundly iff a distributive law exists, and the obstruction to that law is a computable cohomology class. For the one member of the family that has been worked through completely, that class is not only computable but *machine-checked* (Theorem 3.2). The impact claim for the grant is therefore sharp and bounded: the framework *predicts a bug class from the absence of a distributive law*, and it has delivered a sorry-free certificate for the canonical member. What it has not yet delivered — and this note is careful to say so — is the same treatment for the state-desynchronisation members.

What a theorem would require. To upgrade Modelling Claim 5.1 from a dictionary to a theorem, for either mapping, one must:

- (1) **Build the directed containers.** Give the two effects (marker/bank; pool/hook) as explicit directed containers — handoff categories whose shapes are the operations and whose positions are the replies — faithful to the system’s actual control and state flow.
- (2) **Identify the shared object and the prescribed factor $\mathcal{D}$.** Pin down the shared value (supply; balances) as an object acted on by both, and the wide subcategory $\mathcal{D}$ against which the product is taken.
- (3) **Verify (L) and (H).** Check that each hom-presheaf is a free right $\mathcal{D}$ -set and that the reply functor is a disjoint union of abelian vertex groups with trivial left action — or, where (H) fails, descend to the nonabelian matched-pair obstruction (open; see below).
- (4) **Compute $[\omega]$ and match the exploit.** Compute the Baues–Wirsching class and show it is nonzero exactly in the vulnerable configuration, with the exploit transaction as its witness.
- (5) **Formalise the finite core.** As with the anchor, machine-check the finite class computation, leaving the category-to-complex reduction as a clearly-marked pen-and-paper input.

Step 3 carries the main theoretical risk: the anchor’s hypothesis (H) — abelian vertex groups, trivial left action — need not hold for richer effects, and when it fails the obstruction becomes a nonabelian matched-pair H^2 of the Rosebrugh–Wood distributive law [5], which is not yet developed in this programme [8]. The honest status is thus: the *frame* is in place and proved on one example; the *extension* is a well-posed research programme with one identified hard step.

Relation to neighbouring methods. Sheaf-theoretic multi-agent work places obstructions of degree H^0 (consensus) and H^1 (identifiability) on the agents’ *communication* graph, asking whether agents currently agree [13]. The container obstruction sits one degree up, in H^2 , on the *handoff* category, and asks the prior question: can the effects be fused into one consistent contract at all? Re-entrancy and state desynchronisation are, in this reading, the smallest places where the answer is no.

7. CONCLUSION

A smart-contract exploit that belongs to no single component is a failure of composition, and the natural home for failures of composition is the obstruction to a distributive law. For re-entrancy this is a theorem with a machine-checked core: the DAO-class bug is the nonzero degree-two class $[\omega] = \varepsilon$, and the joint contract exists iff the worker fixes the shared token. We have argued — as a modelling claim, with the proof obligations itemised — that the state-desynchronisation family catalogued by auditors is the same obstruction for different effects: two representations of a shared value that no distributive law forces to agree. The next step is not rhetorical but computational: build the directed containers for the Provenance marker/bank divergence or a Uniswap-v4 hook pair, verify freeness, and compute the class. If it is nonzero exactly in the vulnerable configuration, a second rung of the ladder becomes a theorem; if the nonabelian case intrudes first, that is itself the sharp open problem the frame has surfaced. Either way the economic reading is the same: compositional correctness is a cohomology class, and it has a price.

ACKNOWLEDGEMENTS

The agent/contract-as-container and orchestration readings are due to Neil Ghani and the Kodamai programme, building on Niu–Spivak’s polynomial-functor framework; the Zappa–Szép-from-distributive-laws construction is Ahman–Uustalu’s. The security disclosures are Trail of Bits’. The author thanks Neil Ghani and Robin Langer.

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KODAMAI

Email address: `neil@kodamai.com`